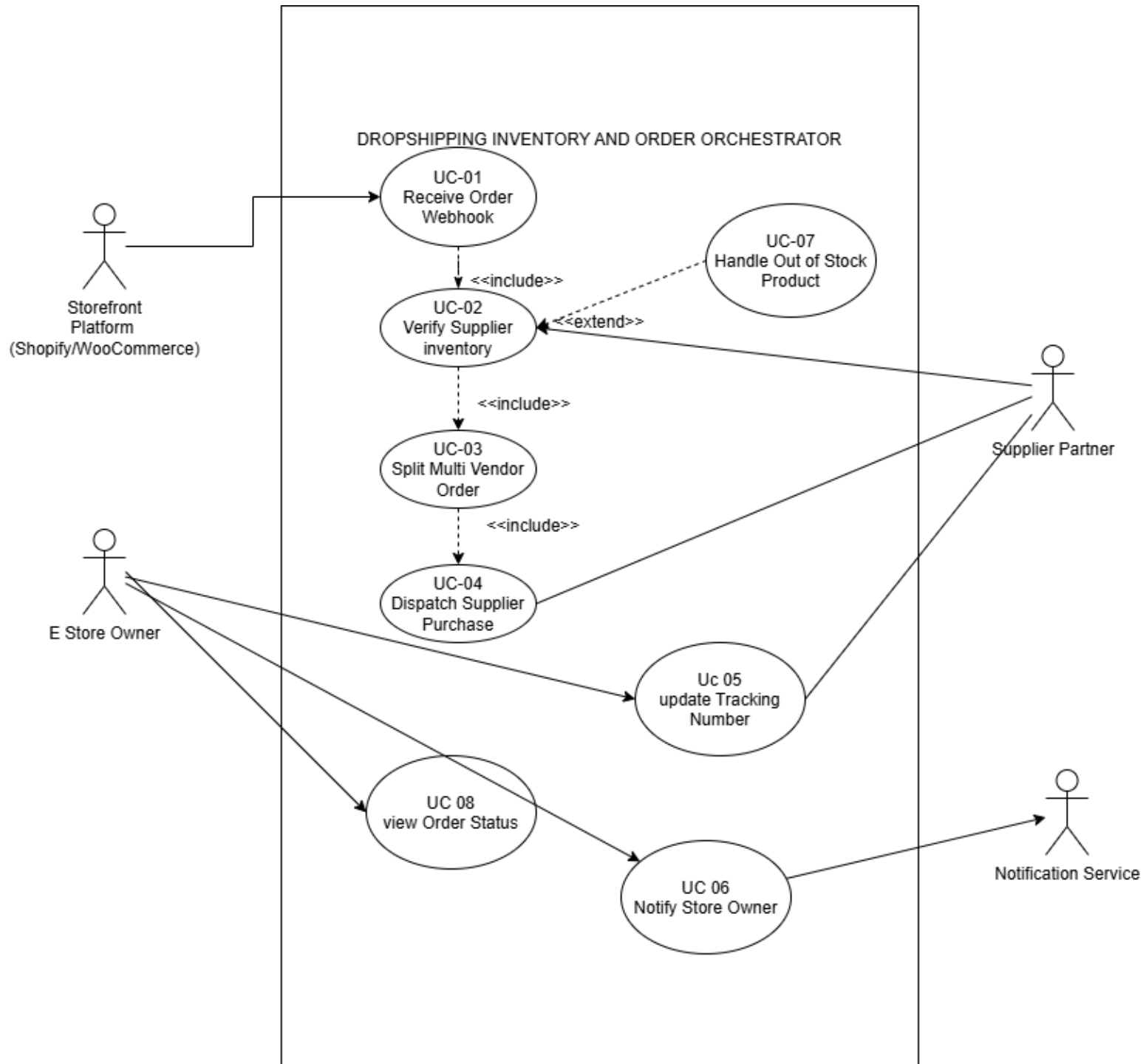


Dropshipping Inventory & Order Orchestrator Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO is triggered with customer delivery address. Fail: Order placed for out-of-stock supplier SKU.	Automates order processing and prevents invalid supplier orders.
FR-002	Functional	The system shall identify the supplier associated with each product in an incoming order.	High	Pass: Every ordered SKU is mapped to its registered supplier. Fail: Any SKU remains without a supplier mapping.	Required to correctly route products to suppliers.
FR-003	Functional	The system shall split a multi-vendor order into separate supplier purchase orders based on supplier ownership of the ordered products.	High	Pass: An order containing products from 2 suppliers produces 2 separate supplier POs.	Allows multiple suppliers to fulfill one customer order.
FR-004	Functional	The system shall record and update the shipment tracking number received from a supplier.	Medium	Pass: A valid tracking number is stored against the corresponding supplier PO and can be retrieved by the store owner.	Enables shipment tracking.
FR-005	Functional	The system shall notify the E-Store Owner when a supplier cannot fulfill an ordered product because it is out of stock.	High	Pass: Owner receives an out-of-stock notification identifying the affected SKU and supplier.	Allows the owner to take corrective action.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storms.	High	Pass: Benchmarking confirms HTTP 200 acknowledgement within 200 ms under simulated peak load.	Prevents duplicate webhook retries and improves reliability.
NFR-002	Nonfunctional	The system shall protect customer and supplier information during data transmission using secure communication.	High	Pass: All external API communication uses HTTPS/TLS. Fail: Any customer/supplier data is transmitted over unencrypted HTTP.	Protects sensitive order and supplier information.



1. Actors

E-Store Owner – views order/fulfillment status and receives notifications.

Supplier Partner – provides inventory availability, fulfills purchase orders, and provides tracking numbers.

Storefront Platform (Shopify/WooCommerce) – sends customer order webhooks.

Notification Service – delivers notifications to the E-Store Owner.

2. Use Cases

ID	Use Case	Description
UC-01	Receive Order Webhook	Receives a new order from the storefront.
UC-02	Verify Supplier Inventory	Checks supplier API inventory.
UC-03	Split Multi-Vendor Order	Separates an order by supplier.
UC-04	Dispatch Supplier Purchase Order	Sends purchase orders to suppliers.
UC-05	Update Tracking Number	Records supplier shipment tracking numbers.
UC-06	Notify Store Owner	Sends order/inventory/shipping notifications.
UC-07	Handle Out-of-Stock Product	Handles unavailable supplier products.
UC-08	View Order Status	Shows current order/fulfillment status to the owner.

3. UML Relationships

UC-01 Receive Order Webhook «include» UC-02 Verify Supplier Inventory.

UC-02 Verify Supplier Inventory «include» UC-03 Split Multi-Vendor Order.

UC-03 Split Multi-Vendor Order «include» UC-04 Dispatch Supplier Purchase Order.

UC-07 Handle Out-of-Stock Product «extend» UC-02 Verify Supplier Inventory.

4. Core Use-Case Flow – UC-01: Receive and Process Order

Preconditions: Storefront integration is configured, supplier information is registered, supplier APIs are available, and a valid order webhook is received.

Main Success Scenario: 1. Storefront sends an order webhook. 2. System validates the webhook. 3. System identifies products/SKUs and suppliers. 4. System verifies supplier inventory. 5. Suppliers return availability. 6. System confirms products are available. 7. System splits the multi-vendor order by supplier. 8. System generates and dispatches supplier purchase orders. 9. Suppliers process orders and provide tracking information. 10. System records tracking/order status for the E-Store Owner.

Alternate Flow – A1: Product Out of Stock: A1. Supplier reports a required SKU is out of stock. A2. System identifies the unavailable SKU and supplier. A3. System does not dispatch an invalid purchase order. A4. System notifies the E-Store Owner and marks the order for further action.

Postconditions: Successful processing creates and dispatches supplier purchase orders and stores order/tracking status. If inventory is unavailable, the affected SKU is identified and the owner is notified.